

DHRUV SOOD

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SUMMARY

AI engineer delivering **production GenAI and ML systems for enterprise clients**: multi-tenant LLM agents with MCP, guardrails and observability, and forecasting and causal models that changed business decisions. Co-author of an LLM alignment preprint.

TECHNICAL SKILLS

LLM Systems: LangGraph, LangChain, MCP, Azure OpenAI, RAG, Multi-Agent Systems, Text2SQL, Tool Calling, Guardrails, Evals
ML & Data Science: pandas, scikit-learn, LightGBM, XGBoost, PyTorch, Forecasting, Causal Inference (DiD), Fine-tuning
Data & Retrieval: PostgreSQL, Redis, OpenSearch, Elasticsearch, Qdrant, Hybrid Search, Apache Airflow, Amazon Redshift
Cloud & Infra: Azure (AKS, OpenAI, ML Registry), AWS, Docker, Kubernetes, FastAPI, Streamlit, Datadog, CI/CD
Programming: Python, SQL, TypeScript, C++

EXPERIENCE

Rocketfrog.ai <i>AI Engineer</i>	Remote Dec 2025 – Present
<i>Client-facing AI consultancy. Delivered production GenAI and ML systems for Sonny’s Enterprises and WashMetrix/Quivio.</i>	
<i>WashMetrix/Quivio — AI analytics chatbot</i> <i>LangGraph, Azure OpenAI, MCP, FastAPI, Redis, AKS, Datadog</i>	
– Built a multi-tenant LLM agent answering business questions with text, charts and tables across 31 sales, labor, membership and accounting KPIs ; served via FastAPI on AKS with Redis-checkpointed state across replicas.	
– Split data access into a standalone MCP service (streamable HTTP); passed tenant scope as request metadata the model cannot alter, with per-request isolation and ownership checks on cached payloads to prevent cross-tenant leaks .	
– Cut LLM cost and latency with a two-tier input guardrail (regex fast path, then LLM scope classifier), a conditional output guardrail and period-aware Redis caching; sandboxed LLM-generated code with a timeout and import allowlist.	
– Enforced schema-validated structured output and enum-typed tool parameters to eliminate invalid model calls; instrumented with Datadog LLM Observability and cross-service trace IDs.	
<i>Sonny’s Enterprises — Site selection & forecasting</i> <i>LightGBM, XGBoost, scikit-learn, Airflow, Postgres, Azure ML</i>	
– Built a site-selection platform forecasting 5-year volume and P&L for any US location (71.7K site-months); LightGBM/ExtraTrees ensemble with opening-ramp and competitor models halved legacy proforma error (29% vs 58% median, p<0.001).	
– Designed a council of 5 LLM agents (LangGraph, GPT-4o, cited web search) recommending whether to build; backtested on 420 historical builds at 52% precision vs a 43% base rate .	
– Ran causal analyses (difference-in-differences, placebo tests, 751 competitor openings) that changed site-selection criteria ; showed 31 demographic measures had no predictive power and cut a claimed campaign lift from +15.9% to +7.1%.	
– Shipped an hourly demand-forecasting service (XGBoost + LightGBM) for 218 sites via Airflow DAGs, an 8.2M-row Postgres warehouse and Azure ML registry; shadow-mode evaluation on 297K site-hours cut MAE 41% (WMAPE 41.5% → 24.5%).	
– Hardened both pipelines with ~50 tests (1e-5 parity tests, regression harness) and a coverage guard that removed a silent failure dropping 158 of 218 sites per run.	
Artificial Intelligence Institute of South Carolina <i>AI Research Intern</i>	Remote Sep 2025 – Dec 2025
– Researched LLM bias, safety, and constitutional alignment frameworks using curated prompt sets, fairness-aware decoding experiments, and cross-model benchmarking across LLMs, achieving up to 26% reduction in identity bias at inference time.	
– Preprint: AMBEDKAR — Fairness-Aware Decoding for Constitutional Alignment.	
Probo (Operations Discontinued) <i>Software Engineer</i>	Gurgaon, India Jan 2025 – Aug 2025
– Led development of a production-grade multi-agent RAG system on OpenSearch, built on a text-to-SQL retrieval pipeline with structured tool calling; balanced quality, latency and inference cost under live traffic, driving a 35% increase in user engagement .	
National Stock Exchange (NSE India) <i>ML Engineering Intern</i>	New Delhi, India May 2024 – Jul 2024
– Architected and deployed a RAG-based regulatory intelligence system over 10K documents on Qdrant using hybrid search.	

EDUCATION & ACHIEVEMENTS

Indraprastha Institute of Information Technology (IIIT-Delhi) <i>Bachelor of Technology in Computer Science & Social Sciences (8.04/10)</i>	New Delhi, India Nov 2021 – May 2025
– Coursework: Machine Learning, NLP, Information Retrieval, Databases, Data Structures & Algorithms, AI Governance	
– Achievements: State Topper (NGSE); Top 5% Adobe GenSolve’24; 2nd place IEEE BlackSlash	
– Leadership: Academic mentor; open-source contributions; technology education initiatives	